

SYNOPSIS

At 1108 on 5 December 2016 the high-speed passenger catamaran *Typhoon Clipper* and the workboat *Alison* collided adjacent to Tower Millennium Pier, River Thames, London. *Alison* capsized and sank immediately; the two crewmen on board were subsequently rescued by *Typhoon Clipper*'s crew. *Alison*'s crewmen were treated at the scene for the symptoms of cold shock, then admitted to hospital for checks before being released later the same day. *Alison* was recovered off the riverbed the following day by the Port of London Authority; there was no pollution.

Alison's crew had unberthed their vessel from Tower Millennium Pier and headed out into the river without properly assessing the shipping situation or making their intentions clear to other vessels. When the crew of *Alison* became aware of the risk of collision with *Typhoon Clipper*, it was too late to take effective avoiding action.

Neither *Typhoon Clipper*'s master nor mate saw *Alison* before the collision. This happened because *Alison* was initially obscured by the pier and then moved into *Typhoon Clipper*'s visual blind sector ahead. *Typhoon Clipper*'s forward-looking closed-circuit television camera captured *Alison*'s movements; however, this image was not being displayed in the wheelhouse.

Neither of *Alison*'s crewmen was wearing a personal flotation device; this placed their lives in immediate danger when immersed in cold water. *Alison*'s owner/operator, Crown River Cruises Limited, had not conducted a risk assessment or developed procedures for the safe operation of its workboats. The investigation also identified ambiguities with the Port of London Authority's regulations regarding the keeping of a lookout on vessels with limited visibility and the use of sound signals when departing from piers.

This report makes a safety recommendation to the Port of London Authority intended to clarify the requirement for keeping lookout on vessels with limited visibility and the use of sound signals when entering the Thames fairway. A safety recommendation has also been made to Crown River Cruises Limited to improve safety through the introduction of workboat operating procedures.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *TYPHOON CLIPPER*, *ALISON* AND ACCIDENT

SHIPS PARTICULARS		
Vessel's name	<i>Typhoon Clipper</i>	<i>Alison</i>
Flag	United Kingdom	United Kingdom
Classification society	Not applicable	Not applicable
IMO number	9451771	Not applicable
Type	High-speed passenger ferry	Workboat
Registered owner	Collins River Enterprises Limited	Crown River Cruises Limited
Manager(s)	MBNA Thames Clippers Limited	Crown River Cruises Limited
Construction	Aluminium	Steel
Year of build	2007	Unknown
Length overall	38.04m	Not applicable
Registered length	35.32m	7.25m
Gross tonnage	169 tonnes	Not applicable
Minimum safe manning	4	1
Maximum number of passengers	220	Not applicable
VOYAGE PARTICULARS		
Port of departure	Tower Millennium Pier	Tower Millennium Pier
Port of arrival	Canary Wharf	Westminster Moorings
Type of voyage	Passenger ferry service	Stores transfer
Passengers	48	None
Crew	4	2
MARINE CASUALTY INFORMATION		
Date and time	5 December 2016, 1108	
Type of marine casualty or incident	Very Serious Marine Casualty	
Location of incident	51°30.42'N - 000°04.69'W	
Place on board	Port hull stem	Hull
Injuries/fatalities	None	One crewman suffered skin burns from contact with red oxide paint
Damage/environmental impact	Indentation to port bow	Vessel foundered

1.2 NARRATIVE

1.2.1 Events prior to the collision

At 0730 on 5 December 2016, *Typhoon Clipper* departed from Woolwich Pier to commence its daily ferry service, initially heading upriver to Westminster (**Figure 1**). After completing one round-trip up and down the river, *Typhoon Clipper* berthed back at Woolwich Pier for a short crew break. During that morning, the crew of the workboat *Alison* took their vessel downriver from St Thomas' moorings and berthed on the north side of Tower Millennium Pier (**Figures 1 and 2**) to collect 10 tins of paint from the offices of Crown River Cruises Limited (CRC).

After the crew rest period, *Typhoon Clipper* commenced its second round-trip service, berthing at Tower Millennium Pier at 1102 (**Figure 2**). At about the same time, the passenger vessel *Silver Bonito* was passing underneath Tower Bridge heading upriver (**Figure 2**).

Once passengers had embarked, the crew of *Typhoon Clipper* unberthed the vessel and the master decided to wait for *Silver Bonito* to pass clear before proceeding ahead. *Typhoon Clipper* was stationary with the bow away from the pier and the port quarter close to, or touching, the pier (**Figure 3**). At the same time, *Alison* was unberthed by its crew and driven close by the end of the pier (**Figure 4**) then headed into the river with the intention of passing ahead of *Typhoon Clipper* before turning to starboard behind *Silver Bonito*.

1.2.2 The collision

At 1107:57 and with *Silver Bonito* passing clear to starboard, *Typhoon Clipper's* master applied ahead power and starboard rudder to head back downriver. Both of *Alison's* crewmen were alerted to *Typhoon Clipper's* acceleration by the sound of its engines increasing power. *Alison's* helmsman immediately applied full astern in an attempt to back out of the way as *Typhoon Clipper* moved rapidly closer (**Figures 5 and 6**).

At 1108:08, *Typhoon Clipper's* port bow struck *Alison's* starboard side abeam the workboat's wheelhouse. *Alison* immediately capsized to port and sank; the crewman on the foredeck was washed overboard and the helmsman was briefly trapped in the wheelhouse before escaping free and swimming back to the surface.

1.2.3 Post-collision events

The master of *Silver Bonito* witnessed the collision and made a "Mayday" report to the Port of London Authority's (PLA) vessel traffic services (VTS) using very high frequency (VHF) radio. PLA VTS responded to the "Mayday" by alerting the Thames RNLI¹ station, local police and ambulance services.

On hearing the impact, *Typhoon Clipper's* master stopped the vessel immediately by selecting full astern. *Typhoon Clipper's* deck/customer services assistant (DCSA) also heard the impact and went to the foredeck to investigate. Hearing shouting and then seeing men in the water, the DCSA reported a man overboard situation to the master using the internal communications system, then threw a lifebuoy (**Figure 7**) that *Alison's* crewmen were able to grab.

¹ Royal National Lifeboat Institution

When the mate heard the DCSA's man overboard report, he went to the port side forward embarkation area, where he used a hand-held radio to inform the master of the situation. Meanwhile, the DCSA threw a second lifebuoy, then rigged the port side scrambling net and fetched the boat hook in preparation for rescuing *Alison's* crew (Figure 8).

Image courtesy of Thames Clippers



Figure 8: *Typhoon Clipper's* port forward embarkation area just prior to the rescue with *Alison's* crewmen in the water

Typhoon Clipper's master manoeuvred the vessel close to *Alison's* crewmen, who were able to cling onto the scrambling net but were unable to climb out of the water unaided. With the assistance of a passenger and another member of Thames Clippers' staff², the mate and DCSA hauled *Alison's* crewmen out of the water, one at a time. The second crewman from *Alison* was lifted out of the water 3 minutes and 30 seconds after the collision (Figure 9).

Typhoon Clipper's master then berthed the vessel back at Tower Millennium Pier, by which time the RNLI, police and ambulance services were in attendance. *Alison's* crewmen were treated for the symptoms of cold shock then taken to hospital; one of *Alison's* crew was also covered in paint that had spilt during the accident. After medical checks in hospital, both of *Alison's* crewmen were discharged later the same day.

² At the time of the accident, there was an off-duty DCSA on board *Typhoon Clipper* who assisted with the rescue of *Alison's* crew from the water, but was not formally allocated crew duties.



Figure 9: *Alison's* second crewman being rescued out of the water by *Typhoon Clipper's* mate, the DCSA, an off-duty DCSA and a passenger

Typhoon Clipper's master was breathalysed by the police after the accident, with a negative result. Due to their evident distress, the crew of *Alison* were not breathalysed, but police records noted that there were no signs that either of them had consumed alcohol.

1.3 ENVIRONMENTAL CONDITIONS

The collision occurred in the sheltered waters of the River Thames during daylight hours with good visibility. The wind was south-westerly and the tidal stream was negligible as the Thames Barrier was closed; the water temperature was 11°C.

1.4 DAMAGE

The stem of *Typhoon Clipper's* port hull was dented where it had struck *Alison* (**Figure 10**). Port of London Marine Services Limited lifted *Alison* off the riverbed the day after the collision (**Figure 11**). *Alison's* wheelhouse superstructure was distorted and the glass from the forward-looking window was missing (**Figure 11 inset**).

rate and a gasp reflex followed by uncontrollable rapid breathing. The onset of the cold shock response peaks within 30 seconds of immersion and lasts for 2 to 3 minutes. If the head is submerged during this time, it can lead to water entering the lungs and potential for death through drowning.

Cold incapacitation usually occurs within 2 to 15 minutes of entering cold water. The blood vessels are constricted as the body tries to preserve heat and protect vital organs. This results in restricted blood flow to the body's extremities, leading to loss of muscle and nerve functions. Useful movement of arms and legs is lost and, unless a lifejacket or PFD is worn, death by drowning can occur due to impaired swimming.

1.12 INTERNATIONAL REGULATIONS FOR PREVENTING COLLISIONS AT SEA

Rule 5 of the COLREGs required every vessel at all times to maintain a proper and effective lookout by sight, hearing and all means available in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision. Furthermore, Rule 7 required every vessel to use all available means appropriate to the prevailing circumstances and conditions to determine if a risk of collision exists.

1.13 ACCIDENT RECONSTRUCTION

On 13 January 2017, with the assistance of Thames Clippers, CRC, LRS and the PLA, the MAIB carried out a reconstruction of the circumstances of the accident using *Typhoon Clipper* and the workboat *Joanna B*. The aims of the reconstruction were to:

- identify if *Alison*'s horn could have been heard by *Typhoon Clipper* master and/or recorded on the wheelhouse audio recorder, and;
- to enable MAIB staff to determine where and when *Alison* could have been seen from *Typhoon Clipper*'s wheelhouse.

Alison was fitted with a single trumpet horn delivering an acoustic power output of 112dB. The specification of the horn on board *Joanna B* could not be identified, but a hand-held horn with an output of 120dB was carried on board *Joanna B* by a member of MAIB staff.

Prior to the reconstruction, it was established that no horn signal was audible on the recording of *Typhoon Clipper*'s wheelhouse audio for the period between *Alison*'s departure from the pier and the collision.

During the reconstruction, *Typhoon Clipper* was positioned adjacent to Tower Millennium Pier and *Joanna B* followed a similar track to *Alison*'s prior to the accident (**Figure 20**). Both *Joanna B*'s fixed horn and the hand-held device were sounded as the workboat departed the pier and again when ahead of *Typhoon Clipper*.

At each stage of the trial, both *Joanna B*'s fixed horn and the hand-held horn were clearly audible in *Typhoon Clipper*'s wheelhouse and, following the reconstruction, could also be heard on the playback of the recording of *Typhoon Clipper*'s wheelhouse audio.

The accident reconstruction demonstrated that, had *Alison*'s horn been sounded at any point between departure from the pier and the collision, it would have been audible on the recording of *Typhoon Clipper*'s wheelhouse audio.



Figure 20: Accident reconstruction - *Typhoon Clipper* bow camera view showing the workboat *Joanna B*

1.14 PREVIOUS OR SIMILAR ACCIDENTS

1.14.1 Collision between *Bowbelle* and *Marchioness*: MAIB Report dated 5 June 1990

On 20 August 1989, the dredger *Bowbelle* and the passenger vessel *Marchioness* collided close to Cannon Street Railway Bridge, River Thames; 51 of the passengers on board *Marchioness* did not survive. The main cause of the collision was that neither vessel was keeping an effective lookout, so neither was aware of the other vessel in time to take avoiding action. The first of the 27 recommendations made in the MAIB report was that, *in all vessels of over 40m in length with a wheelhouse aft navigating in the River Thames above the Thames Barrier, a lookout should be stationed forward at all times.*

1.14.2 Collision between *Brenda Prior* and *Beatrice*: MAIB Report 16/2005

On 17 December 2004, the aggregate carrier *Brenda Prior* and the London Duck Tours amphibious vessel *Beatrice* collided adjacent to Lambeth Bridge, River Thames. Although the lookout, stationed in *Brenda Prior*'s wheelhouse, had reported sighting *Beatrice* prior to the collision, the master had not seen it and took no action to avoid collision.

The MAIB report concluded that *Brenda Prior*'s crew were not keeping an effective lookout. Prior to the accident, the PLA General Direction regarding lookout only applied to vessels over 40m, thus a safety recommendation was made to the PLA to extend this requirement to all vessels with limited visibility.

1.14.3 Collision between *Hurricane Clipper* and *George Williams*

On 20 September 2009, the high-speed passenger ferry *Hurricane Clipper* collided with the traditional Thames cutter *George Williams*. *Hurricane Clipper* had just unberthed from Waterloo Pier when the stem of its port bow struck *George Williams*, breaking the cutter in two. All seven of the cutter's crew went overboard into the river but were rescued by the emergency services. The PLA report into the accident concluded that the accident occurred *because the Master of the HURRICANE CLIPPER did not see the GEORGE WILLIAMS in time to prevent the collision. The combination of the wheelhouse location and the proximity of the cutter to the Clipper made a visual sighting physically impossible.* The report also stated that *Hurricane Clipper* had not made a sound signal in accordance with PLA regulations.

The actions table in the PLA report included the following recommendations:

- *Thames Clippers to change CCTV 4-way split screen display so that port and starboard side are each shown on 2 of the 4-way split screens*
- *Thames Clippers include procedure to ensure that 'off' side is visually checked clear before departing berth*
- *That more small craft be encouraged take steps to make their craft more visible to other users and give a wider berth to larger vessels.*

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE OR ALCOHOL

There is no evidence that the master of *Typhoon Clipper* or either crewman on board *Alison* was suffering from the effects of fatigue or alcohol and, therefore, they are not considered to have been contributing factors to this accident.

2.3 THE COLLISION

Both vessels had an obligation under the COLREGs to assess the risk of collision by keeping a good lookout and evaluating the shipping situation.

Alison's crew did not effectively assess the shipping situation after departing from Tower Millennium Pier and before starting their passage upriver. Although aware of *Typhoon Clipper's* presence, they had not noticed that it had unberthed and was, therefore, likely to accelerate ahead. Thus, the risk of collision with *Typhoon Clipper* went unnoticed by *Alison's* crew until it was too late to take effective avoiding action.

On board *Typhoon Clipper*, neither the master nor the mate saw *Alison* prior to the collision. Initially the pier obstructed their view, and when *Alison* cleared the end of the pier it moved into *Typhoon Clipper's* blind sector ahead. *Typhoon Clipper's* forward-looking CCTV camera did capture *Alison's* movements, but this image was not being displayed in the wheelhouse. Neither vessel made a sound signal to alert others to their intentions.

2.4 ALISON – LOOKOUT AND RISK OF COLLISION

2.4.1 Crew competence

Both *Alison's* crewmen had a great deal of experience working on the river, including on board passenger vessels, where they had developed an innate understanding of blind spots and the associated risk of small vessels not being seen. Both were also qualified as MCA boatmasters with Thames LKEs. Therefore, their decision to take the risk of passing ahead of *Typhoon Clipper* was not a result of insufficient experience or training.

2.4.2 Risk of collision

Having loaded paint, *Alison's* crew unberthed the vessel and proceeded close around the end of the pier, heading for the fairway.

One of *Alison's* crewmen was on the helm and the other was working on the forward deck (**Figure 6**); in this situation, they could easily communicate with one another so were, in effect, both keeping a lookout. Their heads were above the level of the pier (**Figure 21**) and, while their view was slightly obstructed by the pier's railings, they had both seen *Typhoon Clipper*. However, neither had noticed that *Typhoon Clipper*



Figure 21: View of the pier from *Typhoon Clipper*'s wheelhouse (during reconstruction) and inset analysis of height of *Alison* when close by the pier

had unberthed and was likely to accelerate at any moment. They had assumed that *Typhoon Clipper* was still alongside and that it would be safe for them to pass close ahead into the fairway.

As *Alison* continued past the end of the pier, the helmsman was focused on watching *Silver Bonito* pass ahead so that he could time his turn to starboard and head upriver. This distraction, coupled with the assumption that *Typhoon Clipper* was still berthed, led to the risk of collision not being identified on board *Alison*. Therefore, when *Alison*'s helmsman was alerted to the risk, by the roar of *Typhoon Clipper*'s engines, it was too late to take effective avoiding action.

2.4.3 Normalisation of risk

The environment on the River Thames is one where vessels are persistently operating close to each other, particularly smaller craft such as workboats. Thus it becomes routine or normalised for vessels to be frequently managing close quarters situations where risk of collision exists.

To *Alison*'s crew it seemed perfectly normal to be operating very close to two larger vessels, and there was no sense that more time should be taken to assess the situation. This unsafe operating condition was underpinned by an assumption that *Typhoon Clipper* would be able to see *Alison* and would keep clear.

Had *Alison*'s crew stopped to assess the situation before entering the fairway, it would have been clear that trying to pass close ahead of *Typhoon Clipper* was unsafe. There was a hand-held VHF radio on board that could have been used to inform the other vessels of the workboat's plan. It would also have been possible for *Alison* to have stopped before proceeding into the river and, having gained the attention of *Typhoon Clipper*'s master, made a visual signal giving warning of the plan to pass ahead.

2.5 TYPHOON CLIPPER – LOOKOUT AND RISK OF COLLISION

2.5.1 Situation

The location of the accident, where vessels berthing and unberthing at Tower Millennium Pier encountered passing traffic, had been identified in the PLA/ TfL study (Section 1.9.2) as the area of greatest navigational risk on the Thames (**Figure 18**). Given this environment, the ability to see all vessels in close proximity is fundamental for collision avoidance; an issue highlighted by previous accidents where the absence of an effective lookout has been repeatedly identified as the main cause.

Having completed passenger embarkation, *Typhoon Clipper*'s master unberthed the vessel from Tower Millennium Pier. Seeing *Silver Bonito* proceeding upriver, he decided to hold position close to the pier until the intended route was clear. This decision complied with the PLA byelaw requiring vessels entering the fairway to keep clear of others proceeding along the fairway.

2.5.2 Line of sight detection

When *Typhoon Clipper* was waiting for *Silver Bonito* to pass, both the master and mate were in the wheelhouse. As *Alison* unberthed and proceeded around the pier, it would only have been possible for them to have seen the top of its wheelhouse, as the pier itself was obstructing their view (**Figure 21**). However, the master and mate were focused on their intended route, looking ahead and to starboard. Therefore, they were extremely unlikely to catch sight of *Alison* as it passed around the end of the pier.

When *Typhoon Clipper* started accelerating, *Alison* is assessed to have been no more than 15m ahead. With an air draught of 1.7m, no part of *Alison* would have been visible when it was any closer than 21m ahead of *Typhoon Clipper*. To have had a realistic prospect of being seen from *Typhoon Clipper*'s wheelhouse, *Alison* would have needed to be about 50m ahead (**Figure 22**). Therefore, once *Alison* had passed the end of the pier, it entered *Typhoon Clipper*'s visual blind sector ahead and all possibility of a line of sight detection from *Typhoon Clipper*'s wheelhouse was lost.

2.5.3 Use of CCTV system

The primary purpose of *Typhoon Clipper*'s CCTV system was to provide the master with situational awareness of the vessel, specifically to aid ship handling when berthing. When the system was installed, it was not intended as a means of lookout and its value as such was limited at night or in poor visibility.

Use of the CCTV system as an aid to keeping lookout became company policy as a result of the PLA's recommendation, following the *George Williams* accident in 2009, that the *port and starboard side are each shown on 2 of the 4-way split screens* (Section 1.14.3).

As *Alison* passed the end of the pier, it was fleetingly visible on the port side outboard CCTV camera (**Figure 23**), and this was displayed on one of the split-screen quadrants in *Typhoon Clipper's* wheelhouse. However, to have seen this, either the master or mate would have had to be focused on the CCTV display at the exact moment when *Alison* passed through the camera's coverage. This would not have been appropriate action given the priority for keeping a visual lookout.

Post-accident analysis of CCTV recordings showed that the bow camera captured *Alison* directly ahead of *Typhoon Clipper* (**Figure 6**). However, the image from this camera, which provided good coverage of the forward blind spot, was not included in the split screen configuration adopted following the PLA's recommendation.

The COLREGs require use of *all available means* to maintain a good lookout and this, along with the PLA's General Direction 28 reference to *suitable technical arrangements*, could be considered to include the CCTV system. However, the CCTV system had not been designed as an aid to keeping lookout.

Image courtesy of Thames Clippers

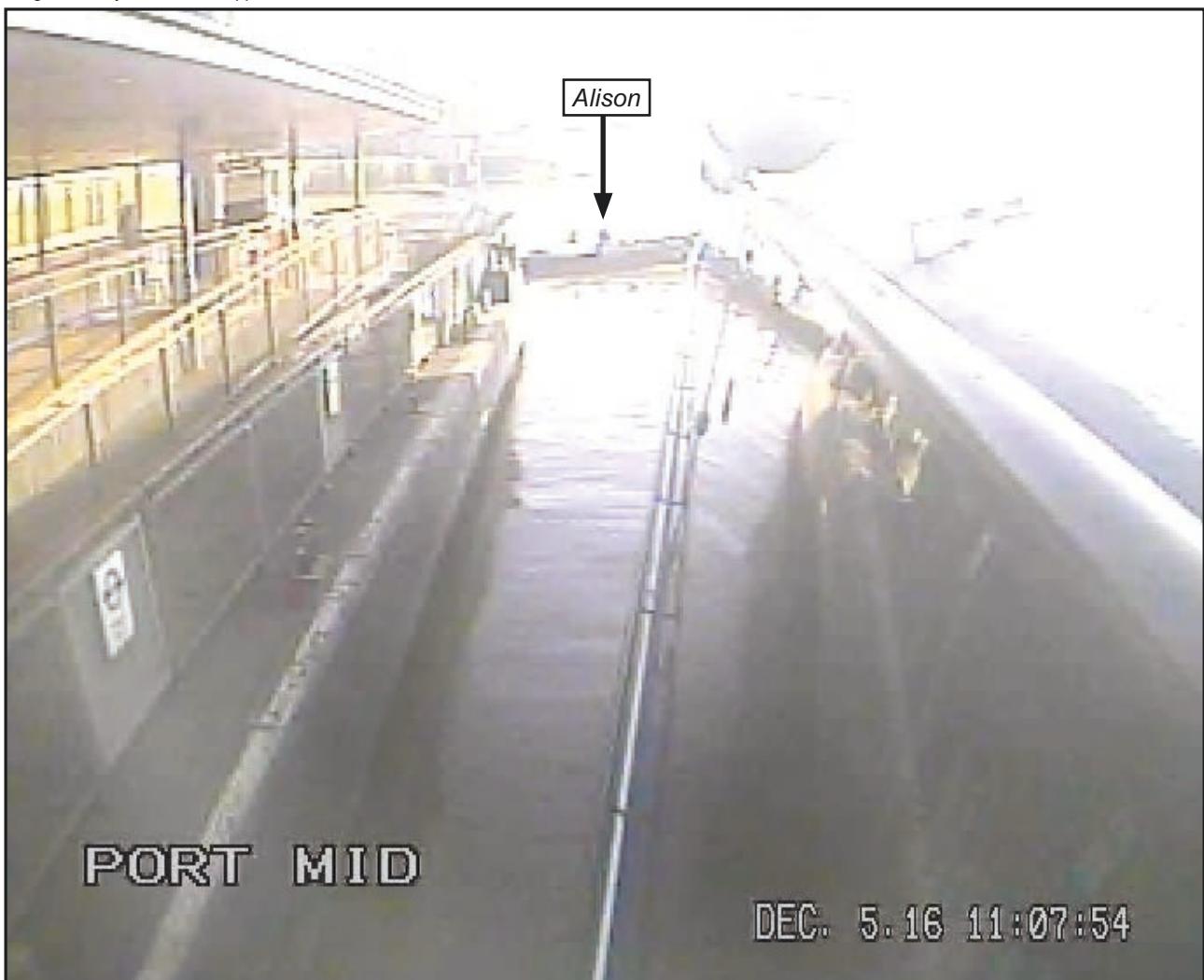


Figure 23: *Typhoon Clipper* CCTV port side outboard forward-looking camera's brief detection of *Alison* immediately prior to collision

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The crew of *Alison* had not effectively assessed the shipping situation and were unaware of the risk of collision with *Typhoon Clipper* until it was too late to take effective avoiding action. [2.4.2, 2.4.3]
2. The helmsman of *Alison* was distracted because he was focusing on *Silver Bonito* passing by. [2.4.2]
3. *Alison's* crew made no attempt to indicate their intentions to other vessels nearby in a busy part of the waterway. [2.4.3]
4. Neither the master nor mate of *Typhoon Clipper* saw *Alison* because it was initially obscured by the pier, then moved into the visual blind sector ahead and the CCTV settings excluded use of the bow camera. [2.5.2, 2.5.3]
5. Neither vessel made a sound signal to warn others of their intentions to enter the fairway. [2.7]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. PLA direction on keeping lookout was ambiguous as it did not define *limited visibility* or *suitable technical arrangements*. [2.6]
2. PLA regulation regarding use of sound signals when entering the fairway lacked clarity as to whether it applied to vessels departing a pier, or not. [2.7]
3. CRC did not have a risk assessment or onboard procedures for safe operation of its workboats, including *Alison*. [2.8.1]
4. The absence of a designated skipper meant that there was a lack of accountability for the safe operation of *Alison*. [2.8.2]
5. *Alison's* crew were not wearing personal flotation devices, an action which significantly reduced their chance of survival when immersed in cold water. [2.10]

3.3 OTHER SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. Although *Typhoon Clipper* had been certified by the MCA as compliant with the HSC Code, the visibility from the vessel's wheelhouse did not meet the Code's requirement. [2.6]
2. The emergency response by *Typhoon Clipper's* crew was swift and effective, ensuring that the crew of *Alison* were rescued quickly. [2.9]